A cheap pre-edit estimate of an edit's reach detects contamination, not propagation

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Abstract

A rank-one weight edit divides its update by a fixed inner product, so the fraction of it arriving at any token can be computed in one forward pass through the *unedited* model — for any prompt, before the edit exists. We ask whether that quantity can screen an evaluation item before the item is run.

It cannot. It is pinned to 1 wherever the prompt names the edited subject, and flat where the question never names it — which is where propagation lives. The same control separates the two regimes by **+0.8439** and by **+0.0002**. What the quantity does detect is contamination, and one published benchmark (Cohen et al., 2024) has items that fail on exactly that ground.

Four further results came off the same probe set. Identity is worth about **sixty-five times** content, which is why there was little for any measure to predict; the two halves of an edit — breaking the old answer and supplying the new one — separate in opposite directions on the same probes; the one universal upper bound on reach, derived in a line from the update's fixed direction, permits a question that never names the edited subject to receive **more** than one that does, and so forbids nothing; and nothing about the subject — neither its prominence nor how many facts depend on it — predicts how hard an edit lands, to within the effect 41 subjects can resolve.

Keywords — model editing, ROME, knowledge propagation, benchmark contamination, negative result

1. Introduction

The task. A weight edit changes one fact and disturbs an unknown amount of everything else. The disturbance is evaluated with hand-written item sets: paraphrases, neighbours, and multi-hop questions whose answers should follow from the edited fact (Cohen et al., 2024), (Zhong et al., 2023). Surveys of the field treat these sets as the measurement (Yao et al., 2023).

Why that cannot bound the damage. An item set gives a lower bound on an edit's damage and never an upper bound. It can show that an edit broke something; it cannot show that an edit broke nothing, because an item nobody wrote and a failure that never happened produce the same output — an empty result. Writing more items does not close the gap: a hand-written set measures the imagination of whoever wrote it, and consistency work shows a model's behaviour varies across surface forms the author did not think to vary (Elazar et al., 2021).

The move that would fix it. Stop enumerating; start measuring. Define a quantity for *any* prompt, so an item can be scored without having been anticipated. ROME offers a natural candidate, because its update is rank one and therefore scalar at every token. The fraction of the update landing on a token is a number, computable in one forward pass through the model as it was **before** the edit.

What this buys, stated narrowly. It removes the need to foresee which items matter. It does not remove the need to have items: the quantity is still evaluated on prompts we hold, and a prompt nobody wrote is still unscored. What changes is that the set no longer has to be *right* in advance, only large.

What this paper finds. The candidate does one job exactly and the intended job not at all. It is pinned to 1 wherever the prompt names the edited subject, which makes it a reliable detector of that naming; and it is flat where the question does not name the subject, which is where propagation would have to be measured. The same algebra that makes it cheap also yields the upper bound an item set cannot give — and that bound turns out to permit a question which never names the subject to receive more than one which does, so it forbids nothing (§5.6). Nor does any property of the edited subject predict how hard an edit lands (§5.7). §5 gives the numbers, §6 the record of how the control that produced them was got wrong once first.

This paper contributes the following.

1. A reach coefficient read off the edit's own rank-one update, computable in one forward pass through the unedited model.
2. A derivation that it is exactly 1 on any prefix-

sharing prompt, with its premise measured on four models from 6B to 405B.
3. Evidence that it does not predict an edit's effect: a raw correlation of rho = 0.522 falls to an excess of 0.085, with an interval touching zero, once a same-subject control is removed.
4. A measurement of why there is little to predict - identity is worth about sixty-five times content - on a pre-stated test.
5. A separation of destruction from destination within one rank-one update, on one probe set under one metric, in opposite directions on the same probes.
6. The measure repurposed as a contamination screen, with one published benchmark scored on it, released as editreach.
7. A universal upper bound on an edit's reach, derived in one line and shown to be vacuous where propagation lives - its median far-field value permits a question that never names the edited subject to receive more than one that does.
8. A second class of candidate eliminated for what sets the magnitude: no property of the edited subject predicts it either, to within the effect 41 subjects can resolve.

2. Related work

The nearest work. He et al. (2026) reports both the depth phenomenon and a cosine decay with layer — 0.80 at layer 5 falling to 0.50 at 25, against 0.98 to 0.47 here — eight months before this one. What it does not carry is a control of the kind in §3.3: its reading of the effect as depth rests on edit success rates staying high across layers, which is a different quantity from the one that would separate a real effect from the measure's own behaviour.

Propagation measured after the fact. Luo et al. (2026) tests how far edits spread against graph structure, finding that *editing tends to over-spread, unintentionally altering related nodes* while unlearning *often under-spreads*, and that *different hierarchical branches exhibit distinct ceilings for achievable accuracy.* The variable it moves is a node's **level** in a hierarchy — root, intermediate, leaf — not the number of facts depending on it, and it uses no measure of how often a subject appears in text. Both of those are relevant to §5.7 and neither is settled by it. Rinberg et al. (2025) produces distance-stratified propagation curves across eight methods, with distance taken from semantic neighbours rather than from an edit's own algebra, and evaluates unlearning rather than editing. Kruthof (2026) labels cascades by dependency, but over manuscripts and prompted editors rather than weights. All three measure outcomes after an edit is applied. None screens an item before it is run, which is the gap this paper tries to fill and then reports it cannot.

Where the edit lands. The MLP layers ROME edits behave as key-value memories (Geva et al., 2021), which is what makes a single projection a sensible place to write a fact. But where a fact can be *located* is not where it can be *edited* (Hase et al., 2023) — so a measure read off the edit's own update, rather than off a localisation, is the one worth trying. MEMIT extends the same rank-one rule to many facts at once (Meng et al., 2023), so the quantity below is defined for it too.

Bounding a perturbation is a field, and it bounds a ball. §5.6 computes an upper bound on how much of a weight edit reaches a token, which is the shape of a certified-robustness guarantee. That literature certifies against an adversary free to choose any direction inside a norm ball: Weng et al. (2020) certifies a region in weight space, handling single-layer perturbation analytically, and Tsai et al. (2021) works from the classification margin. A rank-one edit is a strictly smaller problem — **one** direction, fixed, and known before the edit is applied — and neither paper considers a low-rank perturbation or evaluates a transformer. So weight-space certification is not an open lane and is not claimed to be. What could not be found is certification that exploits a known rank-one direction for editing locality.

Why §5.6's bound stops at the layer it is read in. A bound on the perturbation at one layer says nothing about the model's output without a number saying how much the rest of the network can amplify a change fed into it — a Lipschitz constant. (Kim et al., 2021) is where that number stops being available. It proves dot-product self-attention has no such constant for any p-norm on an **unbounded** input domain, because one term in its derivative equals the sample variance of the input, which has no ceiling. The same paper states that on a *compact* domain it is Lipschitz, and declines that route only because the constant depends on the input range and a tight one would be non-trivial work. The honest position is therefore that **no usable constant exists**, not that no constant exists — a distinction this paper got wrong once before reading the theory sections.

On novelty, plainly. The mechanism in §3 is a line of algebra that any practitioner could derive, and several may assume without writing it down. The claim here is **not** that it is unpublished. It is that it could not be found stated, on one web-search pass and one neighbouring paper read in full — enough to ask the question and not enough to answer it. If it is published, §3 becomes a replication and the contribution is §5.

3. Methods

Everything below is measured on one kind of object. It is worth naming the parts in plain language, with one example carried throughout, before any of them is given a symbol.

The edit. ROME takes a prompt naming somebody — *Henry Mackenzie was born in the city of* — and rewrites one layer of the model so that the next word becomes whatever was asked for. One person, one fact, one layer. That prompt is the **edit prompt**,

and Henry Mackenzie is the example used from here on.

The probe. Any other prompt scored afterwards, to ask how much of that rewrite reached it. A probe opening with the same name — *Henry Mackenzie is the subject here. Henry Mackenzie was born in the city of* — is in the **near field**. A probe that never names him, such as a question about the head of government of the city he died in, is in the **far field**. Multi-hop propagation — the thing an edit is supposed to do and the reason anyone evaluates one — lives in the far field.

The coefficient. For any probe, one number: the fraction of the rewrite that lands on a chosen token. It is read off the arithmetic of the edit rather than by running the edited model, which is why it can be computed before the edit exists. §3.1 gives it.

The floor. A coefficient on its own cannot be judged, because a pipeline hardwired to return 1 looks the same as a correct one. Every reading is paired with a second taken identically but against *somebody else's* edit — the **floor**. If a probe reads the same against a stranger's edit as against its own, the measure has said nothing about this edit.

The run markers. A small marker after a number, like the one at the end of this sentence, names the run that produced it.[E-004] Appendix B lists every run with its model, its size, its date and the directory it wrote. Nothing on this page is a number without one.

3.1 The coefficient

ROME rewrites a single MLP output projection W at one layer so that a chosen key maps to a chosen value (Meng et al., 2022). The rewrite is rank one:

$$W' = W + \frac{(v^* - Wk^*)\,u^\top}{u^\top k^*}$$

For a token whose key at that layer is k, the change to the layer's output is the vector $(v^* - Wk^*)$ scaled by

$$c = \frac{k \cdot u}{u \cdot k^*}$$

so c is computable in one forward pass through the *unedited* model, needing nothing about the edit beyond k^* and u. Appendix A derives the pinning result that follows from it.

3.2 The near field is pinned by arithmetic

Under causal attention, a prompt opening with the edited subject has, at that subject's last token, exactly the text the edit prompt had. Its key is therefore identical to k^*, and $c = 1$ — for any covariance, at any layer, whatever relation the prompt goes on to ask about. This is analytic. What was measured is its premise, and it held exactly: **1.000** across four models from 6B to 405B, served through NDIF (Fiotto-Kaufman et al., 2025). A prompt naming a different person receives **0.082**.[E-018]

3.3 The control, and what it cost to get right

A coefficient of 1.000 means nothing on its own, because a pipeline hardwired to return 1 looks identical. Every reading is therefore paired with a floor: the same prompt scored against a *different* subject's edit. The floor has to match the real edit on everything except the subject — the relation included. Getting that wrong is not hypothetical. It is what this project did first, and §6 keeps the wrong floor on the page as a withdrawn finding rather than deleting it.

3.4 Three arms

Four of the results below are read from the same 495 probes over 42 subjects, scored under three arms:

- **real** — the subject's own edit, applied while probing that subject.
- **same-subject control** — a *different property* of the same person, at comparable magnitude. It holds identity fixed and varies content.
- **different-subject control** — another person's edit, chosen as the nearest edit magnitude. It holds magnitude roughly fixed and varies identity.

Those 495 probes fall into six families, which is how the tables below are broken down. Each asks a different way of reaching the same edited person.

The pair is what makes §5.3 and §5.4 possible: each arm isolates one of the two things a rank-one update carries. Magnitude matching is on `||Δv||/||k*||`, a single scalar, so two edits of equal norm can still differ in direction relative to the probe's key. A directional control is the obvious refinement and is not run here.

Setting. Every measurement uses layer 5, which is where ROME and EasyEdit edit by default. Edits come from CounterFact; questions from MQuAKE-CF-3k (Zhong et al., 2023) and RippleEdits (Cohen et al., 2024). Appendix B lists each run with its model, its size and the directory it wrote.

4. Implementation

`editreach` 0.1.0 is ten modules, about 1,900 lines. Two of its choices are claims rather than tidying.

The coefficient is an object, not an expression. In the predecessor project it was written inline in each run script. It is the thing being claimed, so it has a name and a type.

A reading without a control does not typecheck. The rule in §3.3 was previously stated as prose, and relied on whoever wrote the next script having read it. Here it is `Unbaselined`, and the program stops.

```
$ pip install -e ai/edit-reach
$ python -m editreach.run --config
↪  configs/mquake-far.toml
  model   meta-llama/Llama-3.1-8B (NDIF)
  layer   5     seed 1538     n 900
  → results/2026-11-06-far/  coefficients.parquet
  ↪  config.toml  env.json
$ python -m editreach.report results/2026-11-06-far
  far   own 0.0543  stranger 0.0481  excess +0.0002
```

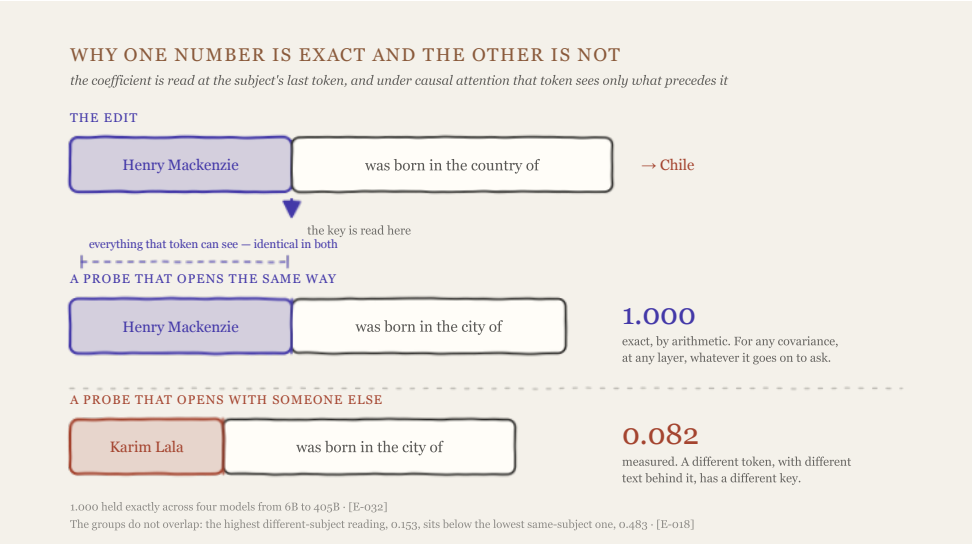


Figure 1: The whole account in one picture. The coefficient is read at the subject's last token, and under causal attention that token sees only what precedes it — so a probe opening with the edited subject carries the same key the edit did, whatever it goes on to ask.

Family	n	what it varies	example prompt
position	210	where the name sits in the prompt, and whether it is there at all	*The city where Henry Mackenzie was born is*
repetition	84	the name appearing more than once	*Henry Mackenzie is the subject here. Henry Mackenzie was born in the city of*
alias	80	a different string for the same person	*Mackenzie was born in the city of*
semantics	42	meaning, with the string held fixed	*It is not true that Henry Mackenzie was born in the city of*
exposure	42	the name present but not the subject of the question	*Karim Lala admired Henry Mackenzie. Karim Lala was born in the city of*
discrimination	37	a different person with a confusable name	*Henry Kissinger was born in the city of*

Table 1: The six probe families, with a real prompt from each. Henry Mackenzie is the edited subject; Karim Lala is an unedited stranger used as a floor.

```
near  own 0.9334  stranger 0.0506  excess +0.8439
Unbaselined: 0 readings
```

The seed and the layer live in the config, not in argv, and every result directory carries the config and the resolved environment beside the numbers.

What it cannot do. Compute a new edit against a remote model. `compute_v` needs a backward pass, which nnsight 0.8.1 refuses mid-trace (Fiotto-Kaufman et al., 2025). The 42 subjects everything here rests on are the ones whose edits were already cached — survivors, not a free draw.

4.1 Whether a remote reading can be read twice

Half the tickets in this project are not about editing at all. They ask whether a number read off a remote model is the same number tomorrow, because a result that cannot be read twice is not a result. Nine decisions came out of that, and the useful shape of them is a split: the **structural** claims are properties of the instrument and hold everywhere tested; every **magnitude** belongs to a deployment.

Structure holds exactly. A ratio whose two terms are read inside one trace came back at **0.000e+00** on every one of fifteen layer-model pairs, across three models, two architectures and depths 0 to 79.[E-011] The suite that checks this runs unchanged on Llama-3.1-8B, Llama-3.1-70B and GPT-J-6B, and carries a per-model table so that a correct deployment running a different model is never reported as a difference.[E-008]

The variation is routing, and it is exact. Readings of one batch through fresh connections par-

tition cleanly by the serving GPU. What first looked like drift of 2.16e-04 is not drift: the quantity takes one of a small number of **exact** values, reproducing bit-for-bit across runs, across nnsight loaders and across days, and which one you get is which replica served the request.[E-012] A daily reading taken two days later returned the same content hash and the same bits as the first. So a comparison between **device-matched** readings needs no tolerance at all; the 2.2e-04 is retained only for readings whose device differs or was never recorded — which is every artifact this project made before 1 November.

```
16 readings · one 126-prompt batch · fresh connection
↪  each · Llama-3.1-8B L5

value               readings   served by
edd2b2b152de2098           8   GPU 0
54a1755349e607dc           8   GPU 2

GPU 0    8 readings   1 distinct value
GPU 2    8 readings   1 distinct value

anchor coefficient (k and k* in one trace)
  deviation from 1.000, both devices ..........
  ↪  0.000e+00
floor median (k* from cache, nothing cancels)
  within a GPU ................................
  ↪  0.000e+00
  across GPUs .................................
  ↪  3.110e-06
```

Each value came from one device and each device returned one value, and the connections alternated rather than arriving in blocks, so it is per-connection routing. The last two lines are the practical content: a ratio read inside one trace cancels the device entirely, which is why every anchor gate in the repository had survived without anyone noticing any of this. That is a property of the gate's design and not of anyone's foresight.

The partition is a property of a deployment on a day. It appeared on 1 November and did not on 3 November, when all sixteen connections landed on one GPU.[E-009] No tolerance may be justified by the partition alone, so every run records how many distinct devices it saw rather than assuming more than one exists.

Batch composition is not free, and that correction was ours. "Batch composition contributes zero" was recorded, generalised into the reference, and then narrowed: it is true of the derived ratio and false of the raw keys, which move by about **2.6% of their magnitude** when the batch is reshaped.[E-007] Any artifact comparing raw activations across runs is bounded there, not at 2.2e-04.

No magnitude holds. The disagreement between two arithmetic sites reads 5.3e-06 on Llama-3.1-8B and 7.5e-09 on GPT-J, which NDIF serves in float32 rather than bfloat16. The 8B and the 70B are one model family and disagree by an order of magnitude, while GPT-J differs because of how it is *served*. So the grouping variable is the deployment — dtype, sharding, tensor shape — not the family. A tolerance written into code is wrong on some other deployment; it has to be measured where it runs.

One claim here was withdrawn by its own follow-up. The arithmetic-site effect was first recorded as growing with depth in bfloat16. A sweep on the 70B at layers 0 to 79 showed it spanning two to three orders of magnitude across depth with **no predictable direction**.[E-010] The instruction that follows changed from "be careful when reading deep" to "measure it at the layer you read", which is a smaller and more useful sentence. The full reference, with the conformance suite that found four errors in it, is in the repository.

4.2 What the record cost

Before any of the above could be trusted, the log files had to be worth keeping. A census found that **124 of 155 committed log files were a single line naming themselves and nothing else** — 80% of the kept record was content-free. The cause was one import: five run scripts import another as a library for its loaders, and that module calls `logs.setup()` at module scope, so every dry-run pass and every run of five other scripts created a log recording nothing. `setup()` now holds the run header and emits it immediately before the first real record, with the file handler opened lazily. Importing a module costs nothing on disk; running it still produces a self-describing log.

5. Results

Five results, in the order they were run. The first two are about the coefficient and are negative. The third and fourth are about what a rank-one update actually does, and are the reason the first two came out the way they did. The fifth applies what survives to a published benchmark.

5.1 The coefficient does not predict an edit's effect

495 probes over 42 subjects, Llama-3.1-8B at layer 5, edits read from cache.[E-001] Against measured movement — the fall in the probability of the pre-edit top-1 answer — the coefficient correlates at $\rho = 0.522$, CI (0.450, 0.587). That number is real and will be quoted, so it needs its account.

A **same-subject control** — a different property of the same person — already accounts for about 72% of the median movement, |movement| 0.1034 against |control| 0.0740. Removing it leaves an excess correlation of **0.085**, CI (−0.003, 0.172), an interval touching zero.

What the quantity actually predicts. Not *this edit reaches this probe*, but *this probe is sensitive to an edit on this subject*. Those are different claims and only the second survives. The question had been written down with both of its answers before the run — if the prediction held, a usable screen; if it did not, the quantity measures delivery and the screen is a proposal — and it closed on the second.

One check worth recording. The `anchor-floor` family — same prompt shape,

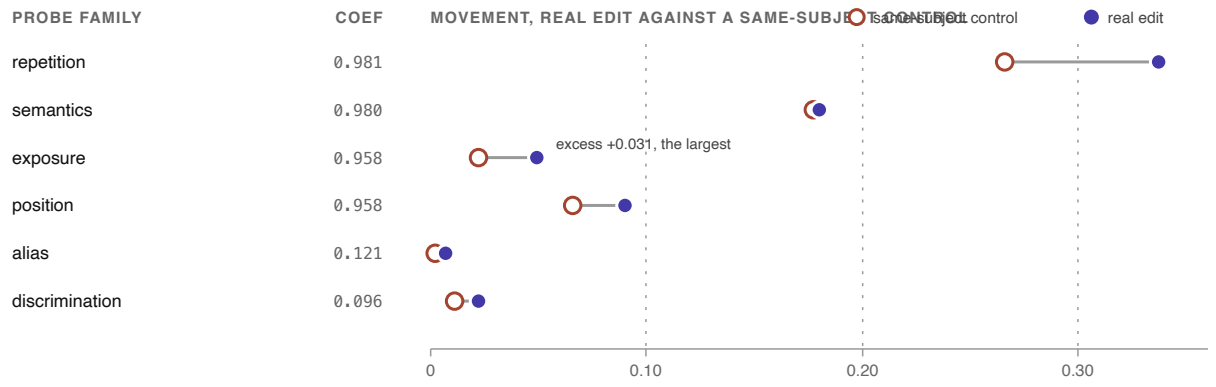


Figure 2: Six probe families, ordered by median coefficient. For each, the movement under the real edit and the movement under a same-subject control, on one scale. Where the two marks nearly coincide the coefficient has predicted nothing the control did not. Only *exposure* opens a visible gap, at +0.031.

Probe family	n	coef med	move med	control med	excess med
alias	80	0.121	0.007	0.002	0.001
discrimination	37	0.096	0.022	0.011	0.003
exposure	42	0.958	0.049	0.022	0.031
position	210	0.958	0.090	0.066	0.001
repetition	84	0.981	0.337	0.266	0.000
semantics	42	0.980	0.180	0.177	0.000

Table 2: Coefficient and movement by probe family. The excess column is what is left after the same-subject control. No family escapes: excess medians run 0.000 to 0.031.

different person — returned a median coefficient of 0.0809 against the independently measured floor of 0.082.[E-018] Two routes to the same number is the cheapest evidence available that the pipeline was measuring what it claimed.

5.2 The far field has no subject-specific dynamic range

Near-field and far-field rows use the same code path and the same floor construction; only the question type differs. The far-field excess is more than three orders of magnitude below the near-field excess, and that gap is the result.

Reading it plainly. In the far field, a stranger's edit reaches a question as much as the real subject's edit does, to within two ten-thousandths. The far field is where multi-hop propagation lives. At layer 5 this measure has no subject-specific dynamic range there at all.

5.3 Identity is worth about sixty-five times content

The same 495 probes, scored under a third arm: another person's edit, matched on magnitude, applied while probing the first.[E-002] Paired differences, bootstrap 10,000 resamples, seed 1538.

The paired differences: real minus same-subject is **+0.0009**, CI (+0.0004, +0.0024) — the contribution of content. Same-subject minus different-subject is **+0.0583**, CI (+0.0472, +0.0732) — the contribution of identity. Both exclude zero, so this is not *content is irrelevant*; it is *content is detectable and is roughly 1.5% of the effect of identity.*

This is why §5.1 came out the way it did. The coefficient fails to predict a particular edit's effect because there is very little edit-specific effect to predict.

A statistic that nearly misled, recorded for that reason. The per-arm medians are 0.0977 and 0.0625, which reads as the real edit being 56% above its same-subject control — a content effect worth more than half the total. But that is a difference of medians across skewed distributions. The arms share their probes, so the estimator that answers the question is the paired median of differences, and it is +0.0009. **The unpaired form overstates the content effect roughly sixtyfold**, which is the difference between a headline and a footnote. Report paired quantities for paired designs.

5.4 Destruction and destination are separable

One rank-one update does two things. It **breaks** the pre-edit answer, and it **supplies** a new one. Measured on the same 495 probes and the same three arms, changing only the readout — from the named token's probability to a full-vocabulary argmax — the two halves separate in opposite directions.[E-003]

```
495 probes · 42 subjects · full-vocabulary argmax ·
↪  Llama-3.1-8B L5
```

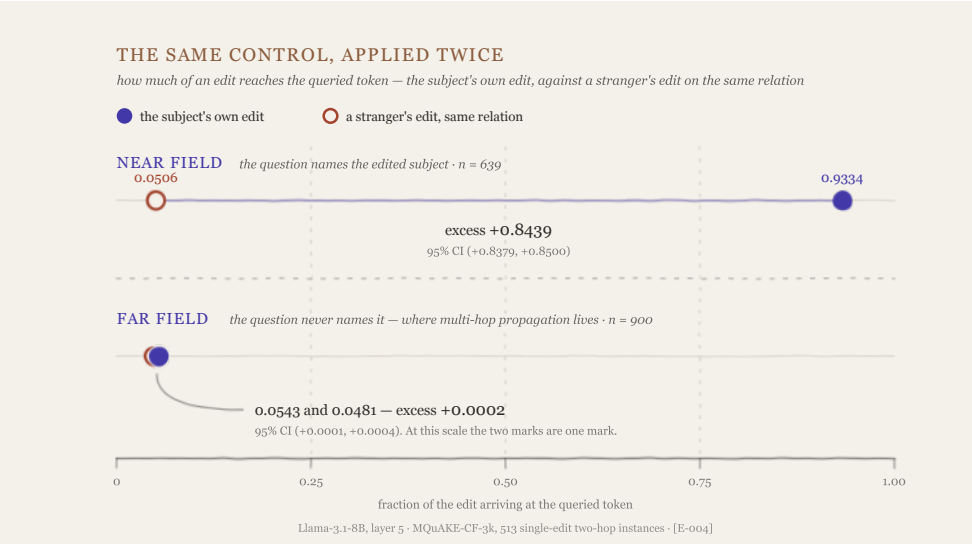


Figure 3: The same control, applied twice. In the near field it separates the real edit from a stranger's by 0.84. In the far field the two marks are one mark. Llama-3.1-8B at layer 5; MQuAKE-CF-3k, 513 single-edit two-hop instances, every hop-2 question scored against all 512 other instances' 'k*', with no edit applied.

Regime	own k*	stranger	excess	95% CI	n	run
Near field — the question names the subject	0.9334	0.0506	**+0.8439**	.8379–.8500	639	E-004
Far field — the question never names it	0.0543	0.0481	**+0.0002**	.0001–.0004	900	E-004
Prefix-sharing premise, four models 6B–405B	1.000	0.082	+0.918	—	16	E-018
Renamed subject, delivered fraction vs. string similarity	$\rho = +0.778$	—	slope +0.847	—	370	E-026

Table 3: Coefficient under the same-relation stranger control, Llama-3.1-8B at layer 5. Intervals are bootstrap percentile.

	base	real	ctl	other
top-1 lands in the injected country	1%	29%	1%	1%

The control and the stranger both sit at the base rate. The real edit is the only arm that puts the new answer anywhere.

Destruction is **subject-keyed**: it follows whose key the edit touched, and the same-subject control reproduces almost all of it. Destination is **content-specific**: it follows what the edit said, and the same-subject control lands in the injected country at 1%, which is the base rate — it delivers nothing rather than a little. Discordant pairs for destination, real against the same-subject control, run 142 to 2.

Which control is the right one depends on which half you are measuring. That is the practical consequence, and it is easy to get wrong in exactly one direction: a same-subject control is at floor for a destination measure and reproduces 98.5% of a destruction measure. Using it on the second is using a control that has already absorbed the signal.

What this does not establish. The control edit injects an occupation while these probes ask for a city, so there is no arm-symmetric check that *each* edit delivers *its own* content. The claim is that content which can answer the question is what arrives — not that every edit delivers its content somewhere.

5.5 What this says about a published benchmark

RippleEdits (Cohen et al., 2024) has a test type meant to check whether an edit propagates to a *different* entity reached through the edited one. Its prompts read *"the name of the head of government of the place of death of **Leo Arons** is"* — and the edited subject is

Probe family	real	same-subject	different-subject
all	**0.0977**	**0.0625**	**0.0016**
alias	0.0065	0.0020	0.0001
discrimination	0.0223	0.0114	0.0067
exposure	0.0489	0.0217	−0.0005
position	0.0900	0.0659	0.0037
repetition	0.3373	0.2656	−0.0076
semantics	0.1795	0.1767	0.0070

Table 4: Median movement by arm, and the paired differences between them. Changing whose key the edit touches is worth about sixty-five times changing what it says.

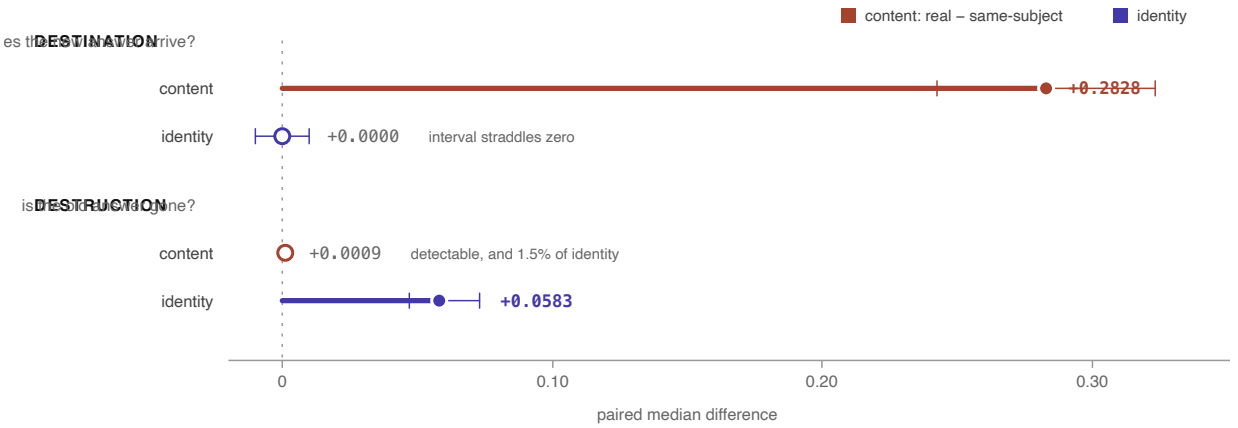


Figure 4: The two halves of one rank-one update, as paired differences on a shared scale. **Content** is the real edit against a different edit on the *same* person; **identity** is that person against someone else. Destination is carried by content and identity contributes nothing; destruction is the reverse. Same 495 probes, bootstrap 10,000, seed 1538.

sitting inside the prompt. Those items receive a median **0.982** of the update against a **0.065** floor built from a stranger's edit on the same prompts.[E-038]

An item like that can be answered by the update arriving at the mention directly, so a score on it cannot separate propagation from delivery. An editor could score on that test while generalising nothing.

The fault is diagnosable rather than universal. MQuAKE names the edited subject when and only when the edit sits on the chain's first hop, which means a clean subset already exists in the field's own data, by construction rather than by design.

5.6 A universal upper bound exists, and it forbids nothing

§1 said an item set gives a lower bound on damage and never an upper bound. That is true of item sets. It is not true of the algebra, and the difference is one line.

Because the update's direction is fixed and only c varies with the token, c is a projection coefficient, and Cauchy–Schwarz bounds it for **every** prompt — no enumeration, no forward pass through the edited model:

$$|c(k)| \le \frac{\|k\|}{\|k^*\|} \qquad \text{with equality exactly when } k \parallel k^*$$

So the upper bound that §1 said an item set could never give is available for nothing. The question is whether it forbids anything, which is empirical.[E-015] Looseness is the bound divided by the coefficient actually read; a bound never approached rules nothing out.

The number that carries this is the far field's **median bound of 1.1456** against a median coefficient of 0.0544. The bound permits a question that never names the edited subject to receive *more* of the update than the near field actually receives. It cannot tell the two apart, so it forbids nothing that happens.

The near field's 1.05 is not a second result. There $k \approx k^*$, which is the same prefix-sharing that pins the coefficient at exactly 1 in §3.2, so Cauchy–Schwarz is near-equality by construction. It is reported because it confirms the implementation. Reading it as a positive result would be reporting §3.2 a second time under a new name.

A prediction that was right for the wrong reason. Before the run, the far-field bound was predicted to read about 1 and therefore forbid nothing. It reads 1.15 and forbids nothing. But the reason given was that the expected squared whitened key norm equals the key dimension — which assumes the covariance is the second moment of keys. It is the identity here, so that argument does not apply. The observed value comes from keys having comparable Euclidean norms, a different fact. The number survived; the derivation that would have carried it to

Half	what it asks	real − same-subj	same-subj − diff-subj
Destination	does the new answer arrive?	**+0.2828** (CI (+0.2424, +0.3232))	+0.0000 (CI (−0.0101, +0.0101))
Destruction	is the old answer gone?	+0.0009 (CI (+0.0004, +0.0024))	**+0.0583** (CI (+0.0472, +0.0732))

Table 5: The two halves of one update, on the same probes, paired, bootstrap 10,000, seed 1538. Each control is at floor for one half and carries the effect for the other.

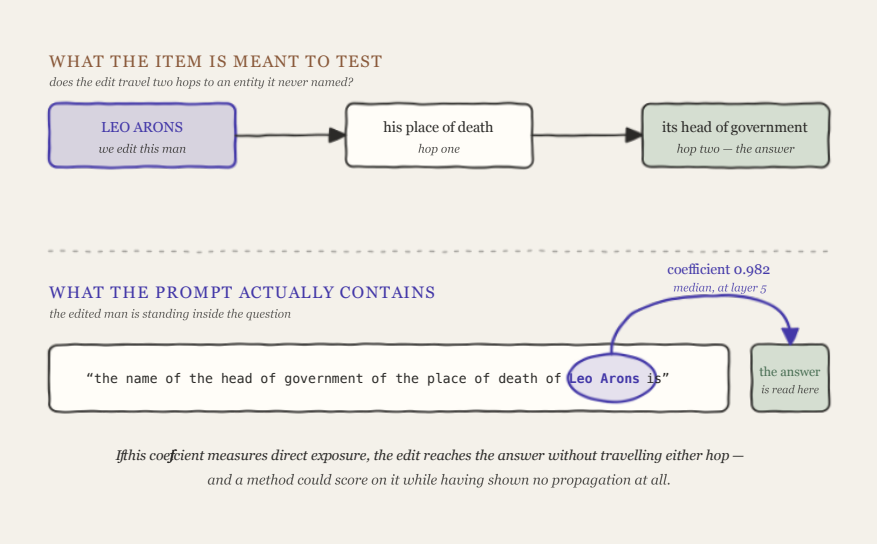


Figure 5: The item is built to ask whether an edit travels two hops. The edited subject is sitting in the prompt, so it does not have to.

another setting did not.

5.7 Nothing about the subject predicts the magnitude either

Seven candidates for what sets the magnitude had already been eliminated, and every one was a property of the activation geometry. This is the first test of a property of the **subject**.[E-016] For each of 41 edited subjects, damage is the median excess — real movement minus the same-subject control of §3.3 — against two subject properties: prominence, as the number of Wikipedia language editions for the subject's Wikidata item, and dependent count, as the number of distinct properties on it.

Every interval spans zero and every point estimate sits within 0.13 of it. Three controls were set in advance and all three came back clean: the two predictors correlate at $\rho = 0.701$, below the 0.8 line at which the partials would have been unstable; controlling for headroom — a better-known subject has more room to fall — moves prominence from +0.059 to +0.038, so there was nothing there to explain away; and recomputing over the four probe families every subject shares leaves prominence at +0.015 and dependent count at −0.039.

What this excludes, and what it does not. At 41 subjects the widest interval spans 0.64, so this rules out a monotone relationship of about $|\rho| > \mathbf{0.37}$ and nothing smaller. It is no evidence of an effect; it

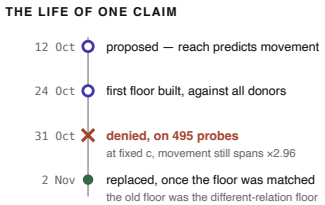


Figure 6: One finding from proposal to withdrawal. The dates stay on the page because a silently corrected claim looks like it was never made.

is not evidence of no effect. A real effect of 0.2 would sit inside every interval above and this study would not have seen it.

Prominence is a proxy and is labelled as one. The question asks about how often a subject's *name* appears in text. Wikipedia language-edition count measures how well an entity is documented, which is a different thing that correlates with it. Its range is real — 0 for a subject with no Wikipedia article at all, 67 for the best covered — so it is not degenerate, but it is not the quantity named.

6. Findings and claims

A paper states what its author believes now. A page revised in place has to state what they believed before and why it changed, otherwise a reader who meets the old claim elsewhere has no way to know it was retired.

Benchmark	Item class	names subject	median c	floor
RippleEdits	built to test two-hop propagation	yes	0.982	0.065
MQuAKE-CF-3k	two-hop chain, edit at the head	6,861 / 6,861	0.933	0.051
MQuAKE-CF-3k	two-hop chain, edit downstream	0 / 16,506	0.054	0.048

Table 6: Direct exposure of the edited subject in two multi-hop benchmarks, counted over the released sets.

Regime	median	c		median bound	median looseness
near field — the question names the subject	0.9334	0.9881	**1.05**	1.05–1.06	10.6
far field — it does not	0.0544	1.1456	**21.12**	20.16–22.14	1715.3

Table 7: How far the bound sits above the quantity it bounds, by regime. 1,539 question-positions over 513 instances, read offline from cached keys.

The table below is every claim this paper has made, numbered in the order they were first written down, each with the observation that would refute it. Struck rows are claims this project withdrew; the second line of such a row says when and why.

7. Limitations

Every qualification in this paper is collected here, so the body above runs unhedged.

1. **Delivery is not effect.** Every number here says how much of the edit vector arrives at a token. None says how far the model's answer moved. At a fixed coefficient, measured movement spans a factor of 2.96.
2. **The coefficient belongs to a token, not a prompt.** Within one prompt the subject position and the final position differ by a factor of 35, and the answer is produced at the final position.
3. **One model, one layer, two-hop chains.** A layer sweep could still find far-field dynamic range at a depth other than 5. It would not rescue layer 5, which is where ROME and EasyEdit edit.
4. **The 42 subjects are not a sample.** They are the cached survivors of a blocked path. Any magnitude here generalises no further than that set.
5. **Nothing here has been read outside the project.** Every finding was produced and reviewed by the same two parties. The self-correction record in §6 is evidence the process works, not evidence the conclusions are right.
6. **A null here is an absence of evidence.** §5.7 rules out a monotone relationship of about $|\rho| > 0.37$ between a subject's properties and the damage an edit does, because 41 subjects cannot see smaller. An effect of 0.2 would be invisible to it. Its prominence variable is also a proxy: Wikipedia language-edition count measures how well an entity is documented, not how often its name appears in text.
7. **§5.6's bound stops at the layer it is read in.** It bounds how much of the update arrives at a token, not how far the output moves. Crossing to behaviour needs a Lipschitz constant for the rest of the network, and §2 reports that none has been computed — not that none exists.
8. **This is not an offer.** The scope above is a finding about a measure, not a method anyone should adopt. Nothing here proposes a better editor, and the novelty position rests on the search described in §2.

8. Conclusion

A quantity computable before an edit exists tells you, reliably, whether a prompt names the edited subject. It tells you nothing about whether an edit travelled — and §5.3 says why there was little to tell: identity is worth about sixty-five times content, so there is very little edit-specific effect for any measure to predict. The two statements are not in tension: the first is why the measure looked promising, the second is why it cannot do the job it was built for. What survives is the first one — a cheap screen that reads an evaluation item and says whether that item can test propagation at all, which one published benchmark fails. Beside it stands a result the screen was not looking for: the two halves of a rank-one update come apart, and which control is correct depends on which half is being measured.

Two negative results were added late and both close doors rather than open them. The upper bound §1 said an item set could never give does exist, in one line of algebra, and it permits a far-field question to receive more of the update than a near-field one actually receives — so the only universal statement available is vacuous where propagation lives. And the search for what *does* set the magnitude, which had eliminated seven geometric candidates, now has a second class of variable in it: nothing about the sub-

Relationship	ρ	95% CI
prominence vs damage	**+0.005**	−0.315, +0.314
dependent count vs damage	**−0.053**	−0.366, +0.270
total statements vs damage	−0.055	−0.370, +0.259
headroom vs damage	−0.130	−0.399, +0.171
prominence, controlling dependent count	+0.059	−0.228, +0.336
dependent count, controlling prominence	−0.079	−0.355, +0.211
prominence, controlling both	+0.038	−0.266, +0.327
dependent count, controlling both	−0.047	−0.340, +0.264

Table 8: Eight relationships, all of them reported. Six were computed, so reporting a subset would be a choice rather than a result.

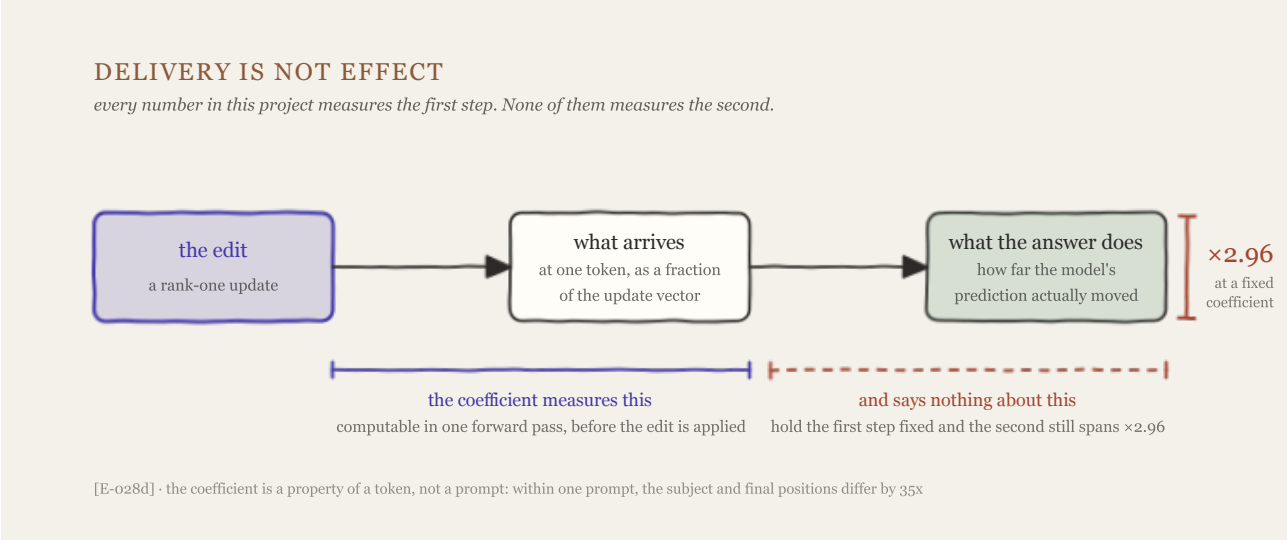


Figure 7: The gap this project existed to close, and did not.

ject predicts it either, to within what 41 subjects can see. **Two classes searched — the geometry of the key and the identity of the subject — and neither carries the magnitude.** That sentence is worth more than the one it replaces, because a reader can tell from it what has *not* been searched.

Five questions stay open. Each is written so it could come back either way, and each names what would change if it did.

1. **Does a layer sweep find far-field dynamic range at some depth other than 5?** If yes, the screen is alive again at that depth and the scope sentence narrows to layer 5. If no, the negative result generalises across depth. This is the cheapest open item: one run on the cached subjects.
2. **Does a graded relation-similarity control move the +0.0002?** If the excess moves with relation distance, the categorical same/different split is too coarse and the far field has structure this one missed.
3. **Is subject-keyed movement the phenomenon rather than the confound?** It surfaced while reading the first run's output, and §5.3 has since measured it on a pre-stated test — identity is worth about sixty-five times content. What stays open is not the measurement but the choice: whether to study the subject-keyed component as the object, or keep controlling it away. *Corrected 9 Nov 2026: this item previously said the question had been deliberately not taken up, to avoid rescuing a withdrawn claim. That reason stopped being true when §5.3 ran, and the item had not been rewritten.*
4. **Does whitening make §5.6's bound informative?** The covariance is the identity throughout this paper. The bound keeps its form under a real covariance but its norm becomes a different quantity, and whitening is precisely the operation that could shrink it in the far field. If it does, the bound forbids something and this section is reopened.
5. **Does the backward-pass block lift?** If nnsight allows `compute_v` mid-trace, the delivery-to-effect gap in §7.1 can be tested on a free draw instead of 42 survivors.

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Every entry above has had its title, authors, year and venue checked against the paper's own record. An entry that has not is drawn with a dotted rule and listed separately on Reading, so the mark appears only where something is wrong — a flag on every line would say nothing.

Appendix A. Why the coefficient is exactly 1 on a prefix-sharing prompt

Let the edit prompt be the token sequence $s_1 \dots s_m$, where s_m is the subject's last token, and let k^* be the layer-ℓ key read at position m. Let a probe be $s_1 \dots s_m\, t_1 \dots t_n$ — the same prefix, then any continuation.

Under causal attention the hidden state at position m is a function only of positions $1 \dots m$. The continuation occupies positions $m+1 \dots m+n$ and therefore cannot enter it. So the probe's key at position m is $k = k^*$ exactly, and substituting into the coefficient:

$$c = \frac{k^* \cdot u}{u \cdot k^*} = 1$$

The cancellation is independent of u, so it holds for any whitening covariance and at any layer, and independent of $t_1 \dots t_n$, so it holds whatever the probe goes on to ask.

The empirical content is entirely in the premise — that the probe really does share the prefix through s_m — which is what E-017 and E-018 measured. The tokenizer is the one place the premise can fail quietly: a leading space or a different casing produces a different token sequence and therefore a different key, with no error raised anywhere.

Appendix B. Provenance of every number

Each row names the run a number came from. Config files pin the seed and the layer; argv carries only the output path.

Appendix C. Readiness

What a short venue note needs, against what exists today. This is a checklist for the author, not a claim about the work.

#	Claim, and what would refute it	value	status
F-01	A prompt opening with the edited subject has the key the edit had, so c is exactly 1 — any covariance, any layer, any relation asked about. **Analytic.** Refuted by a derivation error, or by its premise — prefix sharing — failing. The premise is what was measured.	1.000	standing
F-02	In the far field, a stranger's edit on the same relation reaches a question as much as the real subject's edit does. **Refuted if** the excess exceeds +0.01 at any layer, on a matched same-relation floor.	+0.0002	standing
F-03	RippleEdits items built to test two-hop propagation name the edited subject, so the update arrives by arithmetic rather than by reasoning. **Refuted if** the median falls below the 0.065 stranger floor on a re-scored sample.	0.982	standing
F-04	The fault is diagnosable: MQuAKE names the subject when and only when the edit sits on hop 1. **Refuted by** one counterexample in either direction on the released set.	0 / 16,506	standing
F-05	The efficacy margin reported for ROME and MEMIT is rank-identical to suppression of the true answer, because both optimisers stop at the same target probability. **Refuted if** Spearman ρ falls below 0.95 on any held-out subject set.	$\rho = 1.000$	standing
F-06	A reading with both terms of a ratio inside one trace is bit-identical across runs, replicas and days. **Refuted by** one non-zero disagreement at any layer-model pair.	0.000e+00	standing
F-10	The coefficient predicts that a probe is sensitive to an edit on its subject, not that a particular edit reaches it. The raw $\rho = 0.522$ falls to 0.085 once the same-subject control is removed, with an interval touching zero. **Refuted if** the excess correlation exceeds 0.5 on any probe set with a same-subject control in place.	$\rho = 0.085$	standing
F-11	Changing whose key an edit touches is worth about sixty-five times changing what the edit says. Both differences exclude zero, so content is detectable and is roughly 1.5% of the effect of identity. **Refuted if** the paired real-minus-same-subject difference exceeds the same-subject-minus-different-subject difference on any comparable set.	×65	standing
F-12	Destruction and destination are separable within one rank-one update: it breaks the pre-edit answer because it touched that subject's key, and supplies the new answer because of what it said. **Refuted if** a readout that can express the control's own content shows the same-subject arm delivering there, or if the two halves stop separating in opposite directions on one probe set.	+0.2828	standing
F-13	The variation between remote readings is routing, not drift. The quantity takes one of a few exact values, reproducing bit-for-bit across runs, loaders and days, and which one you get is which replica served the request. **Refuted by** one device-matched pair of readings that differ in any bit.	bit-identical	standing
F-07	The magnitude of that disagreement is a property of the model family, so a tolerance can be written once per family. **Denied 4 November.** The 8B and 70B are one family and disagree by an order of magnitude; GPT-J differs because it is served in float32. The grouping variable is the deployment.	5.3e-06	denied
F-08	Predicted reach predicts how far the answer moves, so the quantity can gate a propagation evaluation before it runs. **Withdrawn 31 October.** 495 probes, 42 subjects. At a fixed coefficient, measured movement still spans a factor of 2.96.	denied	withdrawn
F-14	The arithmetic-site disagreement grows with depth in bfloat16. **Withdrawn** by its own follow-up: a sweep on the 70B at layers 0 to 79 found the effect spanning two to three orders of magnitude across depth with no predictable direction. The instruction became "measure it at the layer you read".	no direction	withdrawn
F-09	There is real far-field signal: scored against all donors, the subject's own edit reaches its hop-2 question more than a stranger's does. **Superseded 2 November.** Nearly all of the strangers used to build that floor were asked about a different relation, so it was a different-relation floor under another name. Row 2 replaces it.	+0.0232	superseded
F-15	A universal upper bound on an edit's reach exists and costs one line, and its median far-field value of 1.1456 permits a question that never names the edited subject to receive more than the near field actually receives. **Analytic in its first half, empirical in its second.** The bound — the delivered fraction is at most the ratio of the two key norms — is refuted by an error in its derivation or by the update ceasing to be rank one with a fixed direction. Its looseness is refuted if the far-field median falls below 10 on any model-layer pair — at which point the bound would forbid something.	21.12	standing

date	ver	what changed, and what it cost
10 Nov 2026	v0.4	The abstract and the masthead caught up with the body, a day late: the abstract still said "two further results" when there were four, and the masthead still read v0.3 while this table had already turned v0.4. Recorded because a paper whose front matter disagrees with its own revision log is the same fault as a ticket whose status disagrees with its closure note.
9 Nov 2026	v0.4	§5.6 and §5.7 added. The upper bound §1 said an item set could never give turns out to exist in the algebra and to forbid nothing, and no property of the subject predicts the magnitude either. §2 gained the certification and Lipschitz positioning, and a claim that no behavioural bound exists through a transformer was withdrawn after the theory sections were read: a compact domain makes one exist, it is simply uncomputed.
9 Nov 2026	v0.3	Limitations gained the delivery-is-not-effect entry, which §5 had been leaning on without stating.
6 Nov 2026	v0.2	F-07 denied. Every per-family tolerance in §4.1 was removed; the section now says a tolerance must be measured where it runs.
2 Nov 2026	v0.2	F-09 superseded by F-02 once the floor was found to be the different-relation floor under another name. §3.3 rewritten around the matched floor.
31 Oct 2026	v0.1	The project's own question closed on its negative branch. The paper's original claim — that the quantity gates a propagation evaluation — was withdrawn and the paper re-scoped to a negative result.

Table 10: Revisions. A number that moves appears here before it appears anywhere else.

id	what it measured	model	n	date	output
E-004	Near and far field under the same-relation stranger floor	Llama-3.1-8B	1,539	6 Nov	2026-11-06-far/
E-001	Predicted reach against measured movement, with a same-subject control	Llama-3.1-8B	495	31 Oct	E-001-predicted-vs-measured.json
E-002	The different-subject arm: identity against content	Llama-3.1-8B	495	31 Oct	E-002-different-subject-arm.json
E-003	The same probes under a full-vocabulary argmax readout	Llama-3.1-8B	495	1 Nov	E-003-destination.json
E-007	Batch reshaping against raw key magnitude	Llama-3.1-8B	8	2 Nov	2026-11-02-shape/
E-008	The conformance suite across two architectures	3 models	36	3 Nov	2026-11-03-models/
E-009	Whether the GPU partition reappears on another day	2 models	32	3 Nov	2026-11-03-gpu/
E-010	Five layers on each of two models, six checks each	2 models	90	3 Nov	2026-11-03-depth/
E-012	Every recorded reading of one quantity, by value and device	Llama-3.1-8B	census	3 Nov	2026-11-03-census/
E-011	Arithmetic-site disagreement across fifteen layer-model pairs	3 models	15	4 Nov	2026-11-04-ndif/
E-017	Prefix-sharing premise, first pass	GPT-J-6B	16	21 Oct	2026-10-21-pin/
E-018	Premise at scale; different-subject floor 0.082	4 models	16	23 Oct	2026-10-23-pin4/
E-026	Renamed subject: string similarity against delivered fraction	Llama-3.1-8B	370	28 Oct	2026-10-28-alias/
E-038	RippleEdits re-run, direct exposure counted	Llama-3.1-8B	23,505	1 Nov	2026-11-01-ripple/
E-015	Looseness of the Cauchy–Schwarz bound, by regime — offline over cached keys	Llama-3.1-8B	1,539	9 Nov	E-015-bound-looseness.json
E-016	Subject prominence and dependent count against per-subject damage	Llama-3.1-8B	41	9 Nov	E-016-subject-properties.json

Table 11: The runs behind the figures and tables above. Directories are under `results/`.

	requirement	state today	where
yes	A result that could have come out the other way	The floor was fixed before the run, and the question closed against the project	§5
yes	One figure that carries the argument	The near-and-far-field figure does the work §5 would otherwise do in prose	§5
yes	Reproduction path	Config, seed and resolved environment beside every result directory	§4
yes	Positioning against the nearest work	He et al. (2026) read in full and distinguished; and for §5.6, Weng et al. (2020) and Kim et al. (2021) read in their problem formulations and theory sections rather than their abstracts, which overturned one claim this paper had already written down	§2
no	Read by someone outside the project	Every finding produced and reviewed by the same two parties	§7.5
no	Breadth across depth	One layer. A sweep on the cached subjects is the cheapest fix	§8.1
no	A delivery-to-effect link	Blocked on the backward pass; 42 survivors, not a sample	§7.1

Table 12: Readiness against a short-paper submission.